

● HARD

● ADVANCED MATH

● ANSWER KEY

Advanced Math

02

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1**C****C. II and III only**

Choice C is correct. Subtracting the expression $x - 29$ from both sides of the given equation yields $0 = x - ax - 29 - x - 29$, which can be rewritten as $0 = x - ax - 29 + -1x - 29$. Since the two terms on the right-hand side of this equation have a common factor of $x - 29$, it can be rewritten as $0 = x - 29x - a + -1$, or $0 = x - 29x - a - 1$. Since $x - a - 1$ is equivalent to $x - a + 1$, the equation $0 = x - 29x - a - 1$ can be rewritten as $0 = x - 29x - a + 1$. By the zero product property, it follows that $x - 29 = 0$ or $x - a + 1 = 0$. Adding 29 to both sides of the equation $x - 29 = 0$ yields $x = 29$. Adding $a + 1$ to both sides of the equation $x - a + 1 = 0$ yields $x = a + 1$. Therefore, the two solutions to the given equation are 29 and $a + 1$. Thus, only $a + 1$ and 29, not a , are solutions to the given equation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2

C

C. 20

Choice C is correct. To express the sum of the two rational expressions on the left-hand side of the equation as the single rational expression on the right-hand side of the equation, the expressions on the left-hand side must have the same denominator. Multiplying the first

expression by $\frac{x+5}{x-5}$ results in $\frac{2(x+5)}{(x-2)(x+5)}$, and multiplying the second expression by

$\frac{x-2}{x-2}$ results in $\frac{3(x-2)}{(x-2)(x+5)}$, so the given equation can be rewritten as

$$\frac{2(x+5)}{(x-2)(x+5)} + \frac{3(x-2)}{(x-2)(x+5)} = \frac{rx+t}{(x-2)(x+5)}, \text{ or}$$

$$\frac{2x+10}{(x-2)(x+5)} + \frac{3x-6}{(x-2)(x+5)} = \frac{rx+t}{(x-2)(x+5)}.$$

Since the two rational expressions on the left-hand side of the equation have the same denominator as the rational expression on the right-hand side of the equation, it follows that $(2x+10) + (3x-6) = rx+t$. Combining like terms on the left-hand side yields $5x+4 = rx+t$, so it follows that $r=5$ and $t=4$. Therefore, the value of rt is $(5)(4) = 20$.

Choice A is incorrect and may result from an error when determining the sign of either r or t .

Choice B is incorrect and may result from not distributing the 2 and 3 to their respective terms

in $\frac{2(x+5)}{(x-2)(x+5)} + \frac{3(x-2)}{(x-2)(x+5)} = \frac{rx+t}{(x-2)(x+5)}$. Choice D is incorrect and may result from

a calculation error.

Q3**A****A. -4**

Choice A is correct. The given quadratic function f is in vertex form, $f(x) = (x - h)^2 + k$, where (h, k) is the vertex of the graph of $y = f(x)$ in the xy -plane. Therefore, the vertex of the graph of $y = f(x)$ is $(2, -4)$. In addition, the y -coordinate of the vertex represents either the minimum or maximum value of a quadratic function, depending on whether the graph of the function opens upward or downward. Since the leading coefficient of f (the coefficient of the term $(x - 2)^2$) is 1, which is positive, the graph of $y = f(x)$ opens upward. It follows that at $x = 2$, the minimum value of the function f is -4 .

Choice B is incorrect and may result from making a sign error and from using the x -coordinate of the vertex. Choice C is incorrect and may result from using the x -coordinate of the vertex. Choice D is incorrect and may result from making a sign error.

Q4**29**

The correct answer is 29. The solutions to a quadratic equation in the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be calculated using the quadratic formula, which gives two solutions: $x = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$ and $x = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$. The given equation is in the form $ax^2 + bx + c = 0$, where $a = 1$, $b = 7$, and $c = 5$. It follows that one solution to the given quadratic equation is $x = \frac{-7 + \sqrt{7^2 - 4(1)(5)}}{2(1)}$, or $x = \frac{-7 + \sqrt{49 - 20}}{2}$, which is equivalent to $x = \frac{-7 + \sqrt{29}}{2}$. It's given that one solution to the given equation can be written as $x = \frac{-7 + \sqrt{k}}{2}$, where k is a constant. It follows that the value of k is 29.

Q5**.0177**

Accept also: .0178, 4/225

The correct answer is $\frac{4}{225}$. An expression of the form $\sqrt[k]{a}$, where k is an integer greater than 1 and $a \geq 0$, is equivalent to $a^{\frac{1}{k}}$. Therefore, the given expression, where $n > 1$, is equivalent to $70n^{\frac{1}{5}}70n^{\frac{1}{6}}$. Applying properties of exponents, this expression can be rewritten as $70n^{\frac{1}{5}}70n^{\frac{1}{6} \cdot 2}$, or $70n^{\frac{1}{5}}70n^{\frac{1}{3}}$, which can be rewritten as $70n^{\frac{1}{5} + \frac{1}{3}}$, or $70n^{\frac{8}{15}}$. It's given that the expression $\sqrt[5]{70n} \sqrt[6]{70n}^2$ is equivalent to $70n^{30x}$, where $n > 1$. It follows that $70n^{\frac{8}{15}}$ is equivalent to $70n^{30x}$. Therefore, $\frac{8}{15} = 30x$. Dividing both sides of this equation by 30 yields $\frac{8}{450} = x$, or $\frac{4}{225} = x$. Thus, the value of x for which the given expression is equivalent to $70n^{30x}$, where $n > 1$, is $\frac{4}{225}$. Note that 4/225, .0177, .0178, 0.017, and 0.018 are examples of ways to enter a correct answer.

Q6**C****C. 36**

Choice C is correct. For a function of the form $fx = ar^{\frac{x}{k}}$, where a , r , and k are constants and $r < 1$, the value of fx decreases by $1001 - r\%$ for every increase of x by k . In the given function, $a = 5,470$, $r = 0.64$, and $k = 12$. Therefore, for the given function, the value of fx decreases by $1001 - 0.64\%$, or 36% , for every increase of x by 12. Since fx represents the value, in dollars, of the equipment after x months of use, it follows that the value of the equipment decreases every 12 months by 36% of its value the preceding 12 months. Since there are 12 months in a year, the value of the equipment decreases each year by 36% of its value the preceding year. Thus, the value of p is 36.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7**A****A. -15**

Choice A is correct. It's given that the graphs of the equations in the given system of equations intersect at the point x, y . Therefore, this intersection point is a solution to the given system. The solution can be found by isolating y in each equation. The given equation $8x + y = -11$ can be rewritten to isolate y by subtracting $8x$ from both sides of the equation, which gives $y = -8x - 11$. The given equation $2x^2 = y + 341$ can be rewritten to isolate y by subtracting 341 from both sides of the equation, which gives $2x^2 - 341 = y$. With each equation solved for y , the value of y from one equation can be substituted into the other, which gives $2x^2 - 341 = -8x - 11$. Adding $8x$ and 11 to both sides of this equation results in $2x^2 + 8x - 330 = 0$. Dividing both sides of this equation by 2 results in $x^2 + 4x - 165 = 0$. This equation can be rewritten by factoring the left-hand side, which yields $x + 15x - 11 = 0$. By the zero-product property, if $x + 15x - 11 = 0$, then $x + 15 = 0$, or $x - 11 = 0$. It follows that $x = -15$, or $x = 11$. Since only -15 is given as a choice, a possible value of x is -15.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8**D**

D. $x + 1 - \frac{2}{x-3}$

Choice D is correct. The numerator of the given expression can be rewritten in terms of the denominator, $x - 3$, as follows: $x^2 - 2x - 5 = x^2 - 3x + x - 3 - 2$, which is equivalent to $x(x - 3) + (x - 3) - 2$. So the given expression is equivalent to

$\frac{x(x - 3) + (x - 3) - 2}{x - 3} = \frac{x(x - 3)}{x - 3} + \frac{x - 3}{x - 3} - \frac{2}{x - 3}$. Since the given expression is defined for $x \neq 3$, the expression can be rewritten as $x + 1 - \frac{2}{x - 3}$.

Long division can also be used as an alternate approach. Choices A, B, and C are incorrect and may result from errors made when dividing the two polynomials or making use of structure.

The correct answer is -2,112. It's given that f is a quadratic function. It follows that f can be defined by an equation of the form $fx = ax - h^2 + k$, where a , h , and k are constants. It's also given that the table shows three values of x and their corresponding values of y , where $y = fx + 4$. Substituting $ax - h^2 + k$ for fx in this equation yields $y = ax - h^2 + k + 4$. This equation represents a quadratic relationship between x and y , where $k + 4$ is either the maximum or the minimum value of y , which occurs when $x = h$. For quadratic relationships between x and y , the maximum or minimum value of y occurs at the value of x halfway between any two values of x that have the same corresponding value of y . The table shows that x -values of 21 and 25 correspond to the same y -value, -8. Since 23 is halfway between 21 and 25, the maximum or minimum value of y occurs at an x -value of 23. The table shows that when $x = 23$, $y = 8$. It follows that $h = 23$ and $k + 4 = 8$. Subtracting 4 from both sides of the equation $k + 4 = 8$ yields $k = 4$. Substituting 23 for h and 4 for k in the equation $y = ax - h^2 + k + 4$ yields $y = ax - 23^2 + 4 + 4$, or $y = ax - 23^2 + 8$. The value of a can be found by substituting any x -value and its corresponding y -value for x and y , respectively, in this equation. Substituting 25 for x and -8 for y in this equation yields $-8 = a25 - 23^2 + 8$, or $-8 = a2^2 + 8$. Subtracting 8 from both sides of this equation yields $-16 = a2^2$, or $-16 = 4a$. Dividing both sides of this equation by 4 yields $-4 = a$. Substituting -4 for a , 23 for h , and 4 for k in the equation $fx = ax - h^2 + k$ yields $fx = -4x - 23^2 + 4$. The y -intercept of the graph of $y = fx$ in the xy -plane is the point on the graph where $x = 0$. Substituting 0 for x in the equation $fx = -4x - 23^2 + 4$ yields $f0 = -40 - 23^2 + 4$, or $f0 = -4 \cdot 23^2 + 4$. This is equivalent to $f0 = -2,112$, so the y -intercept of the graph of $y = fx$ in the xy -plane is 0, -2,112. Thus, the y -coordinate of the y -intercept of the graph of $y = fx$ in the xy -plane is -2,112.

Q10

B

B. -50

Choice B is correct. It's given that the equation $-2x^2 + 20x + c = 0$, where c is a constant, has exactly one solution. A quadratic equation of the form $ax^2 + bx + c = 0$ has exactly one solution if and only if its discriminant, $b^2 - 4ac$, is equal to zero. It follows that for the given equation, $a = -2$ and $b = 20$. Substituting -2 for a and 20 for b in $b^2 - 4ac$ yields $20^2 - 4(-2)(c)$, or $400 + 8c$. Since the discriminant must equal zero, it follows that $400 + 8c = 0$. Subtracting 400 from both sides of this equation yields $8c = -400$. Dividing each side of this equation by 8 yields $c = -50$. Therefore, the value of c is -50 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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